

CoRoT-2b

R-1415

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_140805 · created 2026-08-01T05:40:55+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456874.6865 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.02
Transit depth (Rp/Rs)^2	2.85 +/- 0.68 [%]
Orbital Inclination (inc)	86.9 +/- 1.99
Airmass coefficient 1 (a1)	1.066 +/- 0.018
Airmass coefficient 2 (a2)	-0.053 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	None
Optimal Aperture	2.72
Optimal Annulus	7.86
Transit Duration (day)	0.0814 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.02 vs 0.1667 (deviation 0.11 σ)
Duration fit vs published	0.0814 d vs 0.095 d (deviation 2.71 σ)
Mid-time O-C	-6.4 min
Depth SNR	8.4
Residual scatter	1.71 %

Light curve

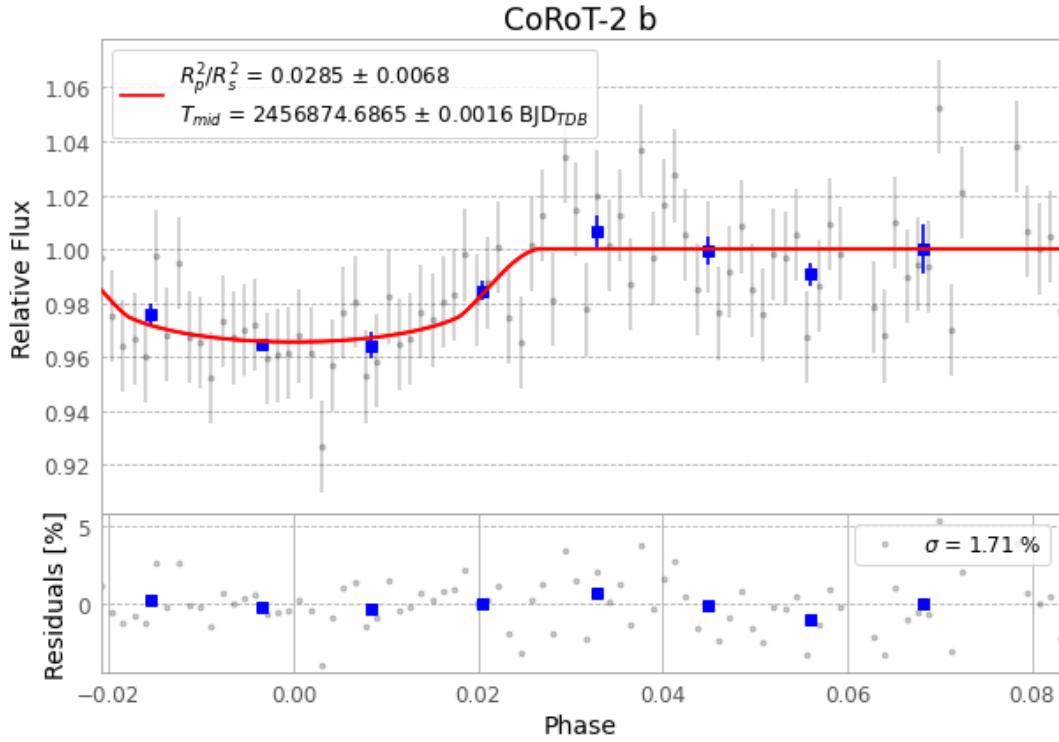


Figure 1 — FinalLightCurve_CoRoT-2 b_2014-08-04.png (SHA-256 f5f5c3c58482e158...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [240, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e402531738f8ec56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), CoRoT-2140805032712.FITS → CoRoT-2140805074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-140805.log (01e813af2d5a...), FinalLightCurve_CoRoT-2 b_2014-08-04.png (f5f5c3c58482...), FinalLightCurve_CoRoT-2 b_2014-08-04.csv (0a6da91d1545...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:40Z.